

Using the hglm package for smoothing noisy data

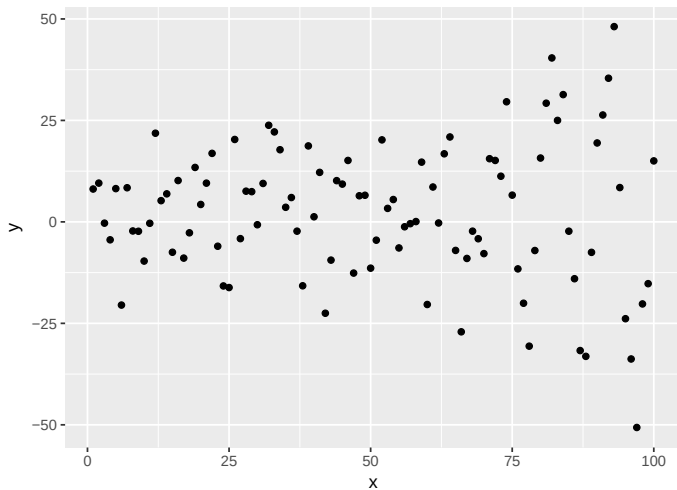
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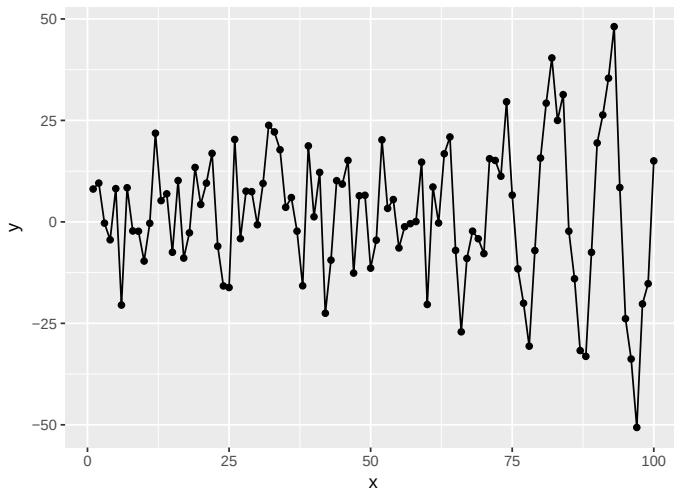
Simulated data

Can you see a pattern??



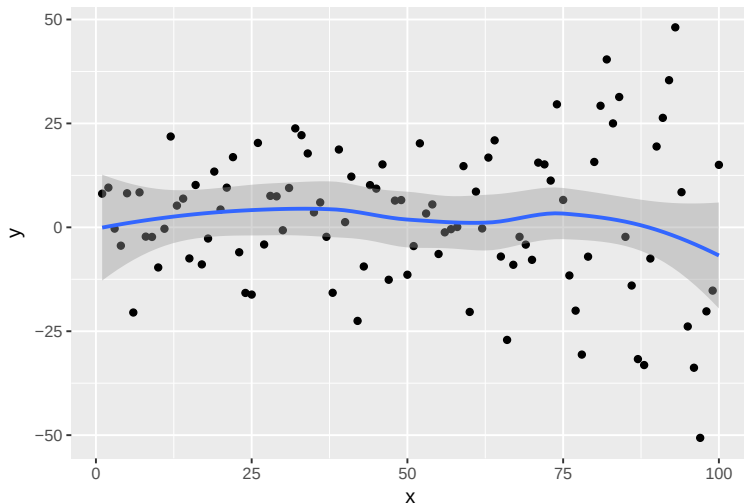
Simulated data

If we join the dots, some kind of oscillating pattern appears.



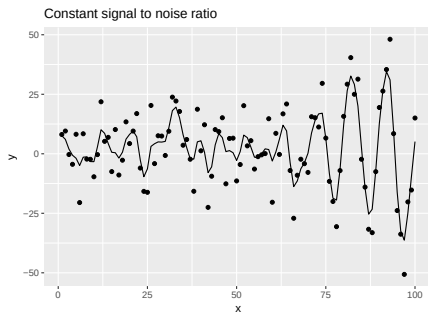
Let's try some standard smoothing methods

Default smoothing ggplot2



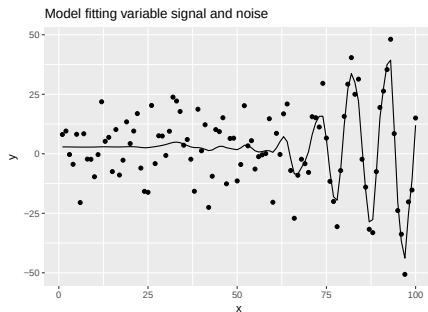
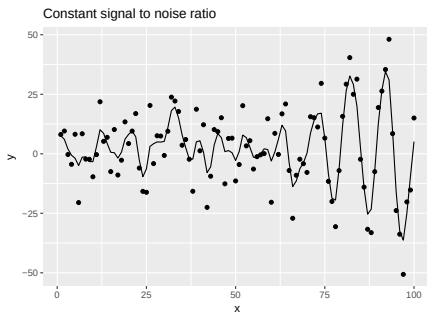
Fitted curve - hglm [Rönnegård et al. 2010]

A Gaussian process can be fitted with a linear mixed model [Pawitan 2001]

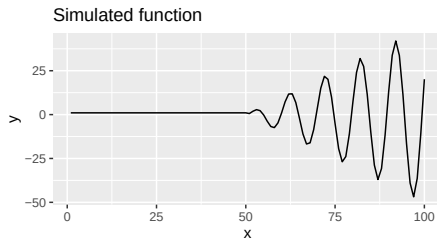


Fitted curve - hglm [Rönnegård et al. 2010]

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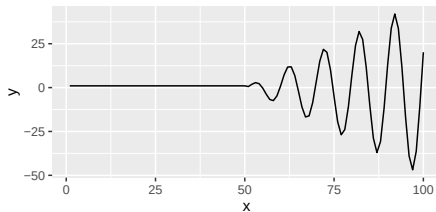


Data simulation

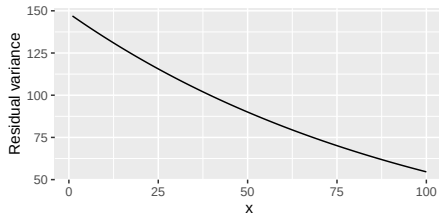


Data simulation

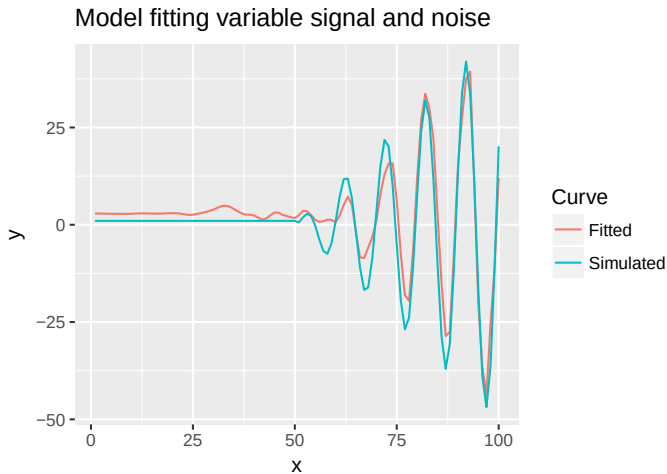
Simulated function



Simulated Residual Variance



Simulated data and fitted curve



Pretty easy to fit using the hglm package in R

Constant noise/signal

```
hglm(y = y, X = X, Z = L )
```

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```

Varying noise and signal

```
hglm(y = y, X = X, Z = L,  
      X.disp = X,  
      X.rand.disp = list(X))
```

References



Pawitan (2001)

In All Likelihood

Oxford University Press.



Rönnegård, Shen, Alam (2010)

hglm: A package for fitting hierarchical generalized linear models

The R Journal 2:20-28.



Lee, Rönnegård, Noh (2017)

Data Analysis Using Hierarchical Generalized Linear Models with R.

Chapman and Hall/CRC